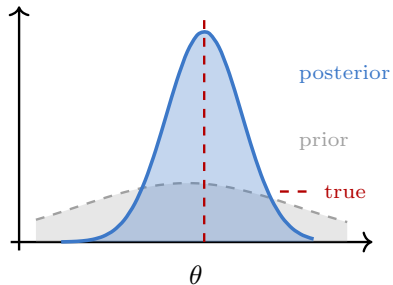


InferCell: Inference-First Whole-Cell Modelling

Tom Kimpson

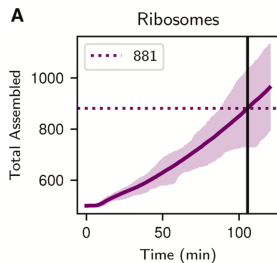
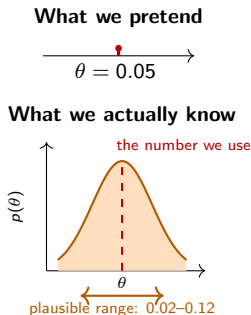
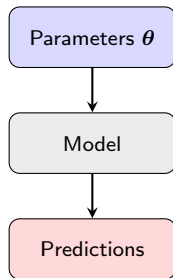
University of Melbourne

MACSYS ISAB Meeting, 12 August 2026



Whole-Cell Models are Forward Simulators

- Representative SOTA: the Luthey-Schulten lab's 4D minimal cell — 493 genes, four coupled solvers, 10 nm voxels, a full cell cycle on GPU ([Thornburg et al., Cell 2026](#)).
- Structurally a **forward simulator**: parameters in, predictions out, no uncertainty propagated.



Thornburg et al., Cell 2026, Fig. 5

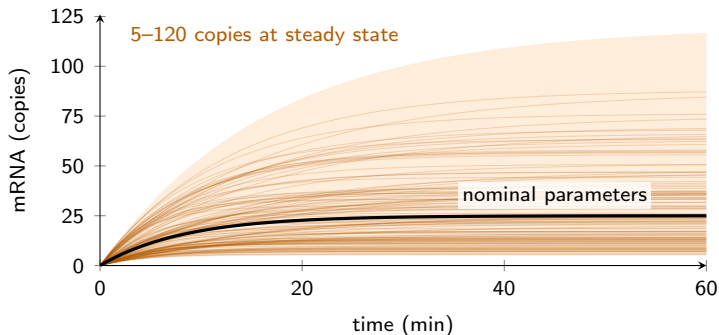
But the Uncertainty Compounds

A minimal model with two free parameters (highlighted):

$$\frac{d[\text{mRNA}]}{dt} = k_{tx} - \gamma [\text{mRNA}] \quad \Rightarrow \quad [\text{mRNA}]_{ss} = k_{tx} / \gamma$$

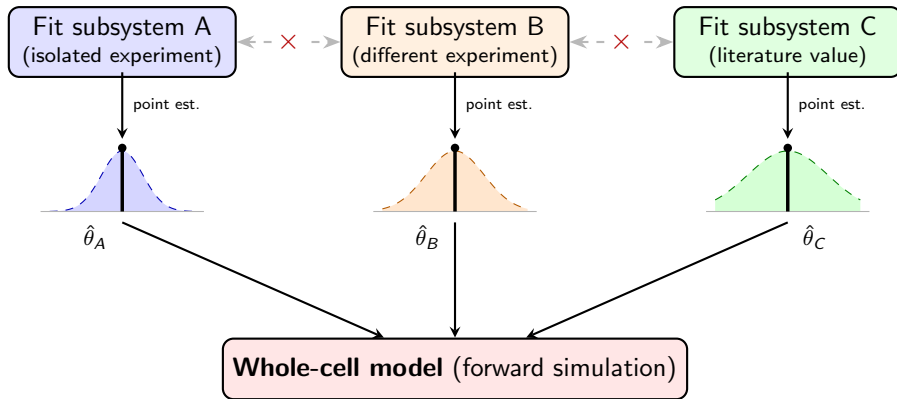
$k_{tx} = 0.05 \text{ s}^{-1}$
measured in *E. coli*, applied to syn3A
range: 0.02–0.12

$\gamma = 0.002 \text{ s}^{-1}$
extrapolated from *M. pneumoniae*
range: 0.001–0.004



No information/uncertainty flows between modules

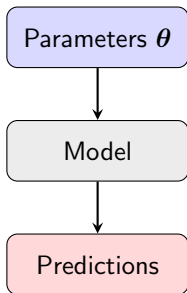
Subsystems are calibrated in isolation, then assembled from point estimates.



Uncertainty is computed, then discarded; the coupling (X) is never seen at all.

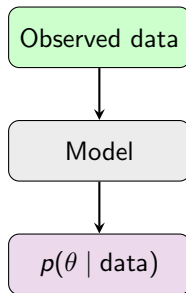
Inverting the Question

Forward simulation



"Given θ , what does the cell do?"

Inference

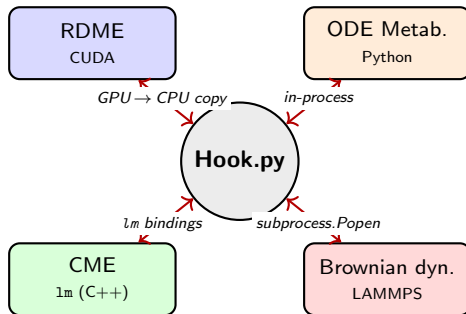


"Given data, what θ are consistent?"

And another thing! - WCMs are polyglot stacks

SOTA 4DMC = four solvers in three languages, glued by a Python script.

- **HPC penalties** — serialise, spawn, copy at every boundary.
- **No differentiability** — AD dies at the boundary.
- **No composability** — every swap is cross-language plumbing.
- **No hybrid inference** — NUTS and SBI, separate ecosystems.
- **Engineering friction** — many toolchains, broken stack traces.



InferCell is a composable, inference-first whole-cell modelling framework in Julia.

github.com/MACSYS-CoE/InferCell

- **Inference-first**

UQ & probabilistic prediction is built in.

Parameters are distributions natively. Boundaries carry posteriors

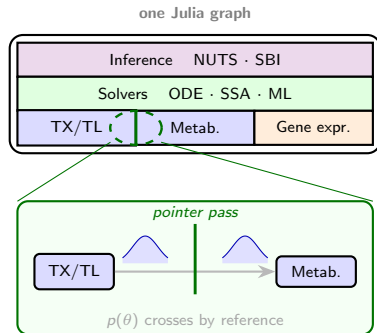
- **Modular.**

Swap ODE for SSA, or a solver for an ML surrogate.

The inference layer is unchanged.

- **Pure Julia.**

One language, one computational graph.



What Becomes Possible (Science)

Questions that forward simulation cannot pose:

- **Probabilistic prediction**

"Is doubling time 110 ± 5 min, or 60–200?"

- **Parameter calibration**

"What is k_{tx} for this gene, given our data?"

- **Identifiability**

"Does data constrain k_{tx} alone, or only k_{tx}/γ ?"

- **Model selection**

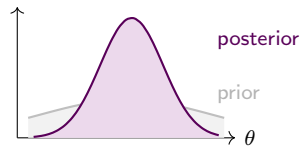
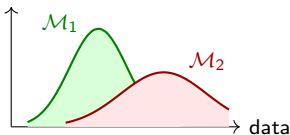
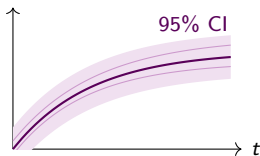
"Does the data prefer ODE or SSA dynamics?"

- **Experimental design**

"Which experiment most reduces uncertainty in γ_{prot} ?"

- **Fundamental biology**

"Does the data require enzyme feedback in metabolism?"



What Becomes Possible (Engineering)

Capabilities that one compiled runtime unlocks:

- **One program**

One type system, one IR, one GPU pipeline

- **End-to-end gradients**

AD through solvers, models and inference:

HMC/NUTS, BOED, Fisher information

- **Hybrid inference**

NUTS and SBI in one pipeline, no
inter-language plumbing

- **Composability**

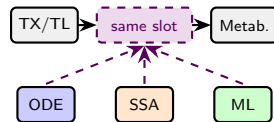
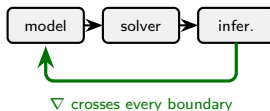
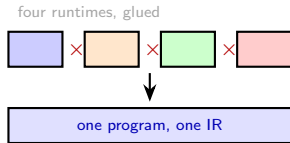
Swap ODE $\leftrightarrow$ SSA, or drop in an ML surrogate
— the inference layer is unchanged

- **Shared infrastructure**

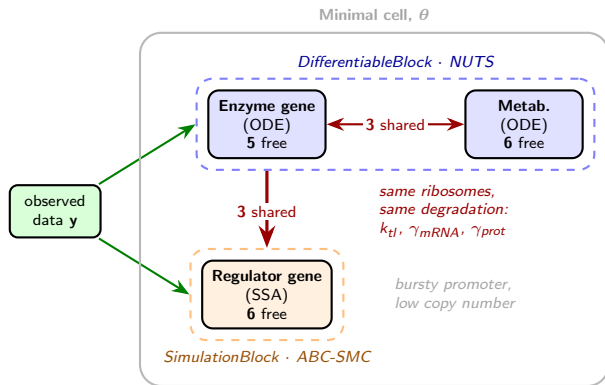
One dependency stack, uniform logging, stack
traces that do not stop at a boundary

- **Cheap prototyping**

Add a module without committing to another
language's ecosystem



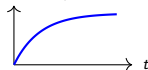
A Minimal Cell, in Three Modules



Goal: compute $p(\theta|y)$

Differentiable dynamics

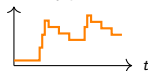
ODE metabolism & expression, continuous kinetics



gradients exist $\Rightarrow$ **NUTS**

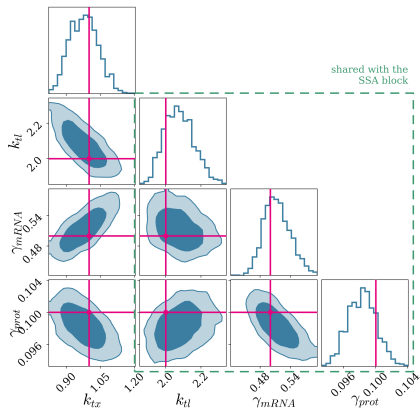
Stochastic dynamics

Gillespie SSA, bursty promoter, discrete counts

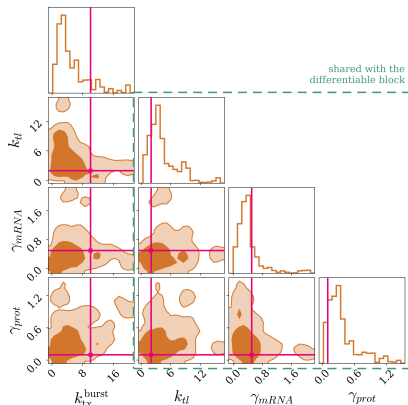


no gradient path $\Rightarrow$ **ABC-SMC**

One Way: Each Block, Its Own Sampler

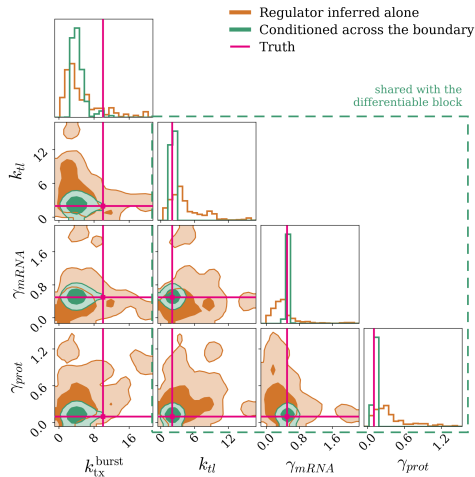


■ **Differentiable block · NUTS**
all 8 parameters jointly, 4 shown



■ **Regulator · ABC-SMC**
all 6 parameters jointly, 4 shown

Another Way: let information flow between modules

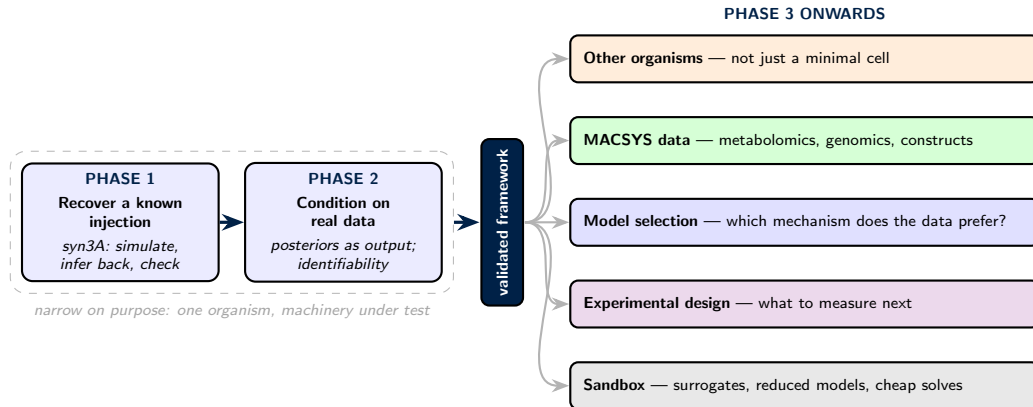


■ **Conditioned** — the shared parameters now carry the differentiable block's posterior across the boundary as their prior.

■ **Alone**, from the previous slide, for comparison.

→ The **3 boxed** parameters sharpen onto truth: exactly the 3 the expression modules share.

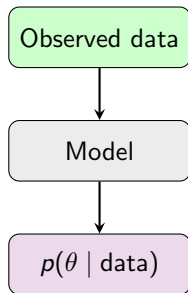
Where This is Going



Summary

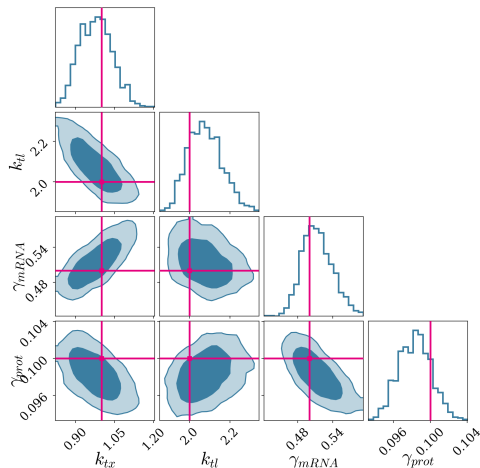
1. **The problem.** Whole-cell models take ~ 1000 parameters from literature and simulate forward. No uncertainty quantification, no way to learn from data, no way to know which predictions to trust.
2. **InferCell inverts the question.** Given observations, infer parameters and their uncertainties — the model becomes something you condition on data, not just run forward.
3. **Toy pilot.** A three-module minimal cell, synthetic data. Information crosses the module boundary and sharpens the posterior onto truth, where the modules share machinery. On synthetic data, so far.
4. **Next steps** Real datasets to condition on; a route into organisms beyond a minimal cell.

github.com/MACSYS-CoE/InferCell



*"Given data,
what θ are consistent?"*

Is the Differentiable Half Identifiable?

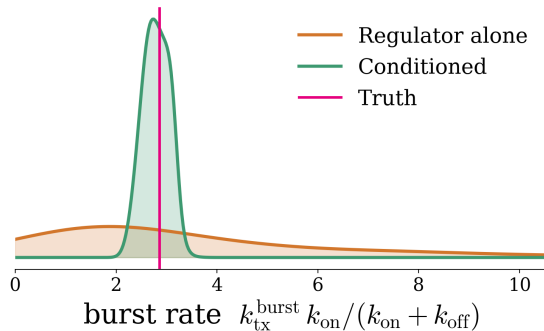


The left-hand posterior from “One Way”, full size: joint NUTS over the differentiable block, all **8** parameters at once. 4 rates shown.

- ■ Magenta lines mark truth.
- **Tilted ellipses** are the point: the data constrains *combinations* of rates, not every rate on its own.

Identifiability measured, rather than assumed — one of the questions forward simulation cannot pose.

Why Do Three Parameters Miss Truth?



The data are replicate-**mean** trajectories. A telegraph promoter enters the mean *only* through the burst rate $k_{tx}^{\text{burst}} k_{\text{on}} / (k_{\text{on}} + k_{\text{off}})$.

So the three rates are **not individually identifiable** from this data — by either block.

Their combination is:

truth	2.86
alone	3.31 ± 2.99
conditioned	2.79 ± 0.24

The boundary sharpens *that* $12\times$ too — and it is why k_{tx}^{burst} 's own marginal tightened $3\times$ without ever being conditioned.

Not a broken sampler. An identifiability result the framework surfaced on its own.

What Would Constrain the Promoter Rates?

Nothing about the model or the sampler — the **summary statistics**.

What we use now

- Replicate-mean trajectories
- Sees the burst rate only
- Identical information content to the mean-field ODE

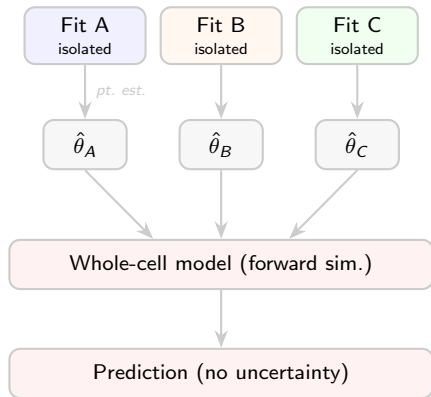
What would separate the rates

- **Fano factor** — bursting pushes it above 1
- **Autocorrelation time** — sets $k_{\text{on}} + k_{\text{off}}$
- **Full count distribution** — higher moments split the burst size from its frequency

If the mean were enough, the mean-field ODE would do. The stochastic module only earns its place once the summaries look past the mean.

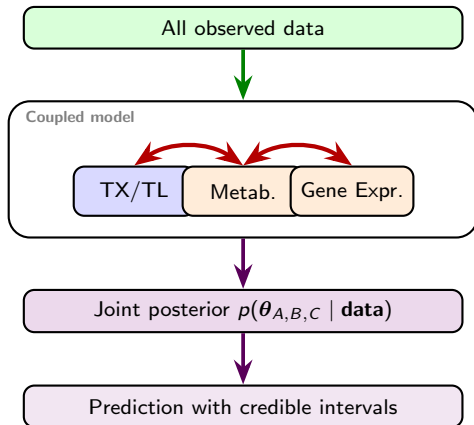
Conditioning the Whole Model

Standard approach



- × uncertainty discarded at every step
- × coupling invisible during fitting

InferCell approach



- ✓ full uncertainty propagated throughout
- ✓ coupling visible during inference

How Much Coupling Do You Need?

Joint inference over all parameters is intractable.

Isolated submodel fits discard all coupling.

